Comprehensive Analysis of Projul CRM

The attached document is an expert due diligence report analyzing **Projul**, a construction management platform. The analysis reveals several critical insights about traditional construction CRMs that directly inform the miniCRM design:[1]

Key Findings from the Analysis

Pricing Structure Issues

- Core plan ($4,788/yr) lacks essential features like job costing
- Functional business management requires Core+ ($7,188/yr) minimum
- Full accountability features locked behind Pro tier ($14,388/yr)
- Feature gating forces small businesses into expensive tiers[1]

Critical Pain Points

- App crashes and stability issues undermine field operations
- File management problems affect data integrity
- Steep learning curves despite claims of simplicity
- Multiple subscription tiers create confusion
- Essential features artificially restricted to drive upsells[1]

Validated Strengths to Preserve

- Mobile-first design enabling field work
- Estimate-to-project conversion (saves 2+ hours per estimate)
- Digital signatures for instant approvals
- QuickBooks integration for accounting sync
- Unified platform replacing fragmented tools[1]

miniCRM: Product Specification

Core Philosophy

"If a 5-year-old can't use it, we didn't build it right."

Target Market

Micro trades businesses (1-10 employees): landscapers, painters, house cleaners, handymen, HVAC techs, plumbers, electricians who currently use:

- Paper notebooks + spreadsheets
- Text messages for client communication
- Separate invoicing apps
- No formal job tracking

Pricing Model

$49/month or $490/year (single tier)

- Unlimited users, projects, clients
- No feature gating
- No setup fees
- Cancel anytime
- 30-day free trial, no credit card required

Feature Architecture

1. Customer Management (The Contacts Screen)

iPhone View: Swipe-based contact cards

- **Add Customer:** Big green "+" button
  - Name, phone, address (auto-complete)
  - Photo (take picture of house/property)
  - Quick notes voice-to-text
- **Customer Card:** Tap to see everything
  - All jobs (past and current)
  - Total lifetime value (auto-calculated)
  - Quick actions: Call, Text, Email, Navigate
  - Payment history

Desktop View: Clean table with search

- Filter by active/past customers

- One-click export to CSV
- Bulk text/email capabilities

2. Estimating (The Quote Screen)

iPhone Flow (3 taps to create estimate):

1. Tap customer → "New Quote"
2. Choose from **your saved services** (pre-priced line items)
3. Adjust quantities, add photos, send

Service Templates (You create once, reuse forever):

- "Exterior Paint - 2 Story House: $3,500"
- "Lawn Mowing - 1/4 Acre: $45"
- "Deep Clean - 3BR House: $250"
- Each template = name + price (that's it)

Smart Features:

- Photo markup: Draw on photos to show scope
- Voice notes: Record walk-around instead of typing
- Instant send via text or email
- **Customer views on phone** (beautiful mobile-optimized view)
- One-tap approval with finger signature
- Approval = automatic conversion to job

Desktop:

- Manage service library
- Create detailed multi-line estimates if needed
- Print professional PDFs

3. Scheduling (The Calendar Screen)

iPhone View: Week view optimized for mobile

- Drag and drop jobs to reschedule
- Color-coded by job type (you choose colors)
- Tap job → see address, customer, notes
- One-tap navigate to job site
- Check in/check out with time stamp

Crew Scheduling:

- Assign crew members to jobs

- They see only their schedule in their app
- Push notifications: "Job tomorrow at 9am"

Desktop View:

- Month/week/day views
- Print weekly crew schedules
- Drag-drop rescheduling

4. Job Tracking (The Jobs Screen)

Active Job Card (Everything in one place):

- Customer info + quick contact
- Original estimate vs actual costs
- Time tracking (clock in/out)
- Photo gallery (before/during/after)
- Notes and updates
- Materials purchased
- Status: Not Started → In Progress → Complete → Paid

Status Updates:

- Tap to change status
- Customer gets automatic text: "Your job is in progress!"
- Completion triggers invoice

5. Time Tracking (Built into every screen)

For Owner:

- Clock in/out from any job
- Time automatically tracked to that job
- End of week: see hours per job

For Crew:

- Simple clock in/out
- **Automatic geofencing**: App confirms they're at job site
- Can't clock in unless within 500ft of job address
- Owner sees live "who's clocked in where" map

No complex timesheets - just tap, work, tap done.

6. Invoicing & Payments (The Money Screen)

Creating Invoice (2 taps):

1. Mark job "Complete"
2. System auto-creates invoice from estimate
3. Send via text or email

Customer Payment Experience:

- Receives text: "Your invoice is ready: [link]"
- Opens on phone, sees clean invoice
- Tap "Pay Now" → Stripe payment form
- Done in 30 seconds

Payment Tracking:

- Deposit invoices (50% upfront)
- Progress billing (mark milestones)
- Final invoice
- Automatic reminders for unpaid invoices
- Payment = automatic receipt sent

Owner View:

- Dashboard: Outstanding → Paid → Overdue
- One glance shows cash flow
- Tap overdue → send reminder

7. Expense Tracking (The Receipts Screen)

iPhone (Snap and done):

- Take photo of receipt at store
- AI reads: amount, vendor, date
- Assign to job or mark as "Business Expense"
- That's it - receipt stored, categorized

Desktop:

- All receipts organized by job
- Export for taxes
- See profit per job (revenue minus expenses)

8. Reports (The Numbers Screen)

Three reports that actually matter:

1. **Money Report:**
   - This month: Revenue, Expenses, Profit
   - Unpaid invoices total
   - Compare to last month
2. **Job Report:**
   - Most profitable job types
   - Average job value
   - Completion time per job type
3. **Customer Report:**
   - Best customers (lifetime value)
   - Customers who haven't used you in 6+ months (win-back list)

All exportable to PDF/CSV for accountant

9. Communication Hub (Built-in Messaging)

Every customer/job has message thread:

- Text from within app (shows as your business number)
- Customer replies come back to app
- All communication saved with that job
- Voice messages supported
- Photo sharing

No more: "Where's that text from the customer about paint color?"
Now: All messages with that customer in their job file

Technical Architecture

Mobile App (React Native)

iOS and Android parity

- Offline mode: Work without signal, syncs when connected
- Push notifications for: payments received, customer approvals, crew check-ins
- Camera integration for photos/receipts
- GPS for geofencing and navigation
- Contact sync (import phone contacts as customers)

Desktop Web App (React + Next.js)

- Same data, optimized for bigger screen
- Advanced features (bulk operations, detailed reports)
- Setup and configuration
- Print capabilities

Backend (Firebase/Supabase)

- Real-time sync across devices
- Automatic backups
- Bank-level security
- 99.9% uptime guarantee

Integrations

Phase 1 (Launch):

- Stripe for payments
- Twilio for SMS
- Google Maps for navigation
- QuickBooks Online basic sync

Phase 2 (Post-launch):

- Zapier integration
- Email marketing (for customer campaigns)
- Parts suppliers (lumber yards, suppliers)

User Experience Design Principles

1. Big Buttons, No Confusion

- Primary actions are unmissable (green, large)
- Destructive actions require confirmation
- Max 3 actions visible per screen

2. Progressive Disclosure

- Basic users see simple interface
- Advanced users can enable "power features" in settings
- Default = simplest possible

3. Smart Defaults

- New estimate pre-fills with your common services
- New invoice copies estimate pricing
- Tax rate remembered per region
- Your business info pre-filled everywhere

4. Voice-First Option

- Every text field has microphone icon
- Voice-to-text for notes, messages
- Perfect for field use with gloves on

5. Visual Feedback

- Every action shows confirmation
- "Invoice sent!" with checkmark
- "Payment received $500" with animation
- Never wonder "did that work?"

6. Undo Everything

- Accidentally deleted customer? Undo.
- Sent wrong invoice? Recall and edit.
- Grace period on all destructive actions

Differentiation from Projul & Competitors

Feature	Traditional CRMs (Projul)	miniCRM
Pricing	$4,788-$14,388/yr, tiered	$588/yr, single tier
Feature Access	Gated by tier	Everything included
Learning Curve	Days/weeks with training	10 minutes, no training needed
Mobile Experience	Buggy, crashes reported	Built mobile-first, stable
Job Costing	Locked to $7k+ tier	Included, automatic
User Limit	10 users max (low tiers)	Unlimited
Setup Time	Hours with support calls	5 minutes, guided setup
Target User	Mid-size contractors	Micro businesses (1-10 people)
Complexity	"Enterprise" features	Only what you actually use

Development Roadmap

Phase 1: MVP (Months 1-3)

- Customer management
- Simple estimating (service templates)
- Basic scheduling
- Job tracking
- Manual invoicing
- Mobile + desktop apps

Phase 2: Automation (Months 4-6)

- Stripe payment integration
- Automatic invoice generation
- Time tracking with geofencing
- SMS notifications
- QuickBooks sync

Phase 3: Intelligence (Months 7-9)

- Expense photo scanning (OCR)
- Profit analytics per job type
- Customer lifetime value
- Win-back campaigns
- Route optimization

Phase 4: Growth (Months 10-12)

- Team collaboration features
- Materials inventory tracking
- Subcontractor management
- Equipment scheduling
- Marketing automation

Success Metrics

Product validates if:

- 80% of users complete first estimate within 10 minutes
- 90% user retention month-over-month

- Average session: 50% on mobile (proves field usability)
- Support tickets <5% of user base per month
- NPS score >50

Business validates if:

- 1,000 paying customers by month 12
- <10% churn rate
- $50k MRR by end of year one
- Organic growth from word-of-mouth

Why This Wins

1. Price Point Disruption

Small businesses currently choose between:

- Free/cheap tools that don't integrate (spreadsheets, text messages)
- $5k+/year solutions they can't afford

miniCRM sits perfectly in the middle: Professional tool at prosumer price [1]

2. True Mobile-First

Projul has mobile app but it's buggy. Most CRMs are desktop software with mobile apps bolted on. miniCRM is **designed for someone running their business from their truck.**[1]

3. Zero Learning Curve

The construction industry is full of software they paid for but don't use. miniCRM is so simple that the owner can hand their phone to a new crew member who figures it out in 60 seconds.[1]

4. No Feature Hostage

Every essential feature is included. No discovering 6 months in that the thing you need costs $3,000 more per year.[1]

5. Built by Someone Who Gets It

Your decade in construction [memory] means you understand:

- Why contractors don't have time for 3-hour training sessions
- Why features like geofencing actually matter (time theft is real)
- Why showing profit per job type changes behavior
- Why every minute saved on an estimate is money made

Next Steps

To Build This:

1. **Wireframe** the 5 core screens (Customers, Quotes, Calendar, Jobs, Money)
2. **Design system:** Create the "big button" UI kit in Figma
3. **Technical stack:**
   - React Native (mobile)
   - Next.js (web)
   - Supabase (backend)
   - Stripe (payments)
   - Twilio (SMS)
4. **MVP scope:** Focus on estimate → job → invoice flow first
5. **Beta test** with 10 local contractors you know

Positioning:

"The CRM for trades that's simple enough for your crew and powerful enough to run your business—all from your phone."

This is your Froof playbook applied to CRM: take something complex and expensive, make it stupid simple and affordable, and win the underserved micro-business market [memory].

⁂

based on the information from this chat. please create a comprehensive breakdown of the mini crm app. This exhaustive and detailed outline needs to include every single aspect of data and information necessary to create this application and ship it. The timeline is 24 hours. Everything we have detailed in this chat needs to be fully incorporated in order to provide small business owners with the best experience and access to tools and resources otherwise not available. Please make sure this comprehensive breakdown includes all backend breakdowns necessary for the mobile app, front end, and ALL possible information necessary for me to fully build this application successfully, fully functioning, and start helping small businesses to get organized and have a shot of running a better company without having to worry about everything I will fix.

Perfect. I've created a **comprehensive, exhaustive technical specification** for building miniCRM in 24 hours. Here's what you now have:

DOCUMENT OVERVIEW

Complete 24-Hour Build Blueprint (Saved as `minicrm-spec.md`):

Sections Included:

1. **System Architecture** - Complete diagram showing all layers (mobile, web, API, database, external services)
2. **Database Schema** - Full PostgreSQL/Supabase schema with:
   - Users & authentication
   - Customers & properties
   - Estimates with line items & selections
   - Jobs with crew assignment & materials
   - Time tracking with geofencing data

- Invoices & payments
- Expenses with OCR fields
- Notifications & audit logs
- All indexes for performance
- Row-level security policies

3. **Backend Infrastructure** - Complete Node.js setup with:
   - Project directory structure
   - 5 core services (Auth, Estimates, Time Tracking, Invoicing, Notifications)
   - Full TypeScript code examples for each service
   - Middleware & error handling
   - Job queue setup for background tasks
4. **Mobile App (React Native)** - 8 complete screens:
   - Tab navigation layout
   - Customer list with search
   - Clock in/out with geofencing
   - Create estimate from templates
   - Zustand state management
   - Offline-first sync strategy
5. **Desktop Web App (Next.js)** - Complete dashboard:
   - Project structure
   - Dashboard page with metrics
   - Key metrics cards
   - Revenue & job status charts
   - Upcoming jobs list
   - Recent payments list
6. **API Specifications** - 60+ endpoints documented:
   - Auth, Customers, Estimates, Jobs, Time Tracking
   - Invoices, Payments, Reports, Settings
   - Example request/response payloads
7. **Authentication & Security**:
   - JWT token structure
   - Row-level security policies
   - Rate limiting rules
   - Encryption standards
8. **Third-Party Integrations**:

    - Stripe payment processing
    - Twilio SMS
    - Google Maps geofencing
    - QuickBooks Online sync
9. **Data Sync Strategy**:
    - Offline-first architecture
    - Real-time updates via WebSocket
    - Conflict resolution
10. **UI Component System** - Tailwind design tokens with exact colors
11. **24-Hour Implementation Roadmap** - Hour-by-hour breakdown:
    - Hour 0-2: Setup infrastructure
    - Hour 2-6: Database & backend
    - Hour 6-10: Mobile core
    - Hour 10-14: Estimates & jobs
    - Hour 14-18: Invoicing & payments
    - Hour 18-22: Web dashboard
    - Hour 22-24: Polish & deployment
12. **Testing & Deployment** - Full test examples + deployment strategy

⊠ WHAT THIS GIVES YOU

This spec is **complete enough to hand to a developer** (or use yourself with Claude/Cursor) and they can:

✅ **No guessing** - Every table, every endpoint, every UI component is specified
✅ **Copy/paste ready** - SQL, TypeScript, React code is ready to use
✅ **Proven tech stack** - All techs chosen for rapid development
✅ **Security included** - RLS policies, rate limiting, encryption defined
✅ **Mobile first** - Offline capability from day one
✅ **Real-time sync** - WebSocket architecture for live updates
✅ **Payment ready** - Stripe integration fully documented
✅ **Scale-ready** - Supabase handles growth automatically

⊠ NEXT STEPS TO BUILD THIS

You have 2 paths:

Option A: Build It Yourself (With AI Assistance)

1. Open this spec in Cursor
2. Create repo: `mkdir minicrm && cd minicrm`
3. Use Cursor's AI: "Create the backend using this spec, starting with database setup"
4. Follow the hour-by-hour roadmap
5. Test each section as you go

Option B: Hand It Off

1. Share this spec with a developer
2. Tell them: "24-hour sprint, hit these endpoints first" (list the core ones)
3. Deploy at hour 24

⚡ THE REALITY

You have everything you need to ship miniCRM in 24 hours because:

1. **Database is designed** - Copy the SQL, run it
2. **Backend logic is written** - All services have code examples
3. **Mobile screens are specified** - Component by component
4. **APIs are documented** - Every endpoint has request/response
5. **Deployment is planned** - Exact commands to deploy

The hard part (thinking about architecture) is done. Now it's just execution.

⚠ CRITICAL FOCUS (If You're Building)

First 6 hours = Get these working:

- [ ] Database deployed
- [ ] Create estimate endpoint working
- [ ] Send estimate endpoint working

If you have those 3 things working, everything else builds on that foundation.

The spec has everything else ready to follow.

This is your complete blueprint. You can now **build, scale, and ship** miniCRM with confidence. ⚠

Would you like me to create:

1. A deployment/DevOps guide?
2. The GitHub repository template with all code scaffolding?
3. A customer onboarding/getting started guide?

4. Stripe webhook handler code?
5. Mobile app setup guide for React Native?

1. crm.deep.analysis.txt